Implementation Plan

6-DOF Robotic Arm Pick-and-Place and Color Sorting

Repository: https://github.com/saumyadebug/robotic-arm-pick-and-place
Source: IMPLEMENTATION_PLAN.md

Implementation Plan · 6-DOF Robotic Arm Pick-and-Place & Color Sorting

Project: Automatic Color-Based Object Sorting Using a 6-DOF Robotic Arm
Repository: https://github.com/saumyadebug/robotic-arm-pick-and-place
Report (poster): `24BRS1075_24BRS1065_24BRS1126_Poster_A3 (7).pdf`
Version: 1.1 (aligned with project report requirements)
Date: 2026-10-22

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1. Purpose

This plan documents the work required to bring the repository into alignment with the project report requirements and to record what has been implemented, changed, added, and removed. The report specifies:

- A **6-DOF robotic arm** actuated by **six MG995 servo motors**.
- Control via **Arduino Uno** + **PCA9685 16-channel 12-bit PWM driver** (I2C).
- Power via a **7.4 V battery** stepped down by an **XL4016 buck converter**.
- A **gripper-mounted TCS3200 color sensor** classifying objects into **four color-coded bins: Red, Green, Blue, Yellow**.
- An **HC-SR04 ultrasonic sensor** for pick-zone detection.
- An autonomous **Finite State Machine (FSM)** control loop.

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2. Gap Analysis (Before)

Requirement (Report)	Repository State (Before)	Gap
6-DOF / 6 MG995 servos	Code used 5 channels (0-4); 6th channel unused	Missing 6th joint (wrist roll)
4 color bins (R/G/B/Y)	FSM classified only R/G/B; 3 bins	Missing Yellow detection & 4th bin
XL4016 buck converter	Generic buck converter documented	Naming/spec mismatch
Gripper-mounted color sensor	TCS3200 documented, not specified as gripper-mounted	Doc gap
Debounced ultrasonic trigger	No debounce in FSM	Robustness gap
Oscillator frequency consistency	Inconsistent across sketches	Reliability gap
Servo channel 5 in BOM/pinout	Marked "spare/optional"	Doc gap
Yellow in color test results	Test log already included Yellow, code did not	Code/test mismatch

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3. Implementation Phases

Phase 1 · Firmware Core (6-DOF + 4-Color FSM)

- Add 6th servo channel `CH_WRIST_ROLL` (channel 5) to all sketches.
- Extend `setJoints()` to actuate 6 joints + gripper.
- Add `COLOR_YELLOW` and 4th bin target (base tick 590).
- Implement Yellow classification in `scanColor()` and `color_sensor_test`.
- Standardize PCA9685 init (`setOscillatorFrequency(27000000)` + `setPWMFreq(50)`).
- Add 3-reading debounce in `STATE_IDLE`.
- Align ultrasonic detection threshold to 10 cm (debounced).

- Add filter-settling delays in `scanColor()`.

Phase 2 · Hardware Documentation Alignment

- Update BOM voltage regulator to XL4016.
- Update power supply guide to XL4016.
- Update system architecture diagram to 6 MG995 joints.
- Update pin connection table channel 5 (Wrist Roll, MG995).
- Mark color sensor as gripper-mounted in sensor integration doc.
- Update development log to 6-DOF.

Phase 3 · Test Records & Status

- Add wrist roll row to servo test results.
- Add Yellow cycle to pick-and-place test results.
- Update README project status and progress checkboxes.

Phase 4 · Planning & Reporting Artifacts

- Create this implementation plan (`IMPLEMENTATION_PLAN.md`).
- Generate change-report PDF (`docs/change_report.pdf`).
- (Optional) Add CI/compile check via Arduino CLI.
- (Optional) Add unit tests for color classification logic.

Phase 5 · Future Improvements (Backlog)

- Implement true trapezoidal/S-curve interpolation in `moveArmSmooth`.
- Implement inverse kinematics for Cartesian positioning.
- Add automated conveyor belt feed.
- Add performance benchmark graphs and demo videos.
- Ambient light shielding & white-balance calibration routine.

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4. Change Summary

4.1 Added

- `CH_WRIST_ROLL` / `CH_WRIST_PITCH` channel definitions (6-DOF).
- `COLOR_YELLOW` enum value and 4th bin (base tick 590).
- Yellow classification logic in `scanColor()` and `color_sensor_test`.
- Ultrasonic debounce counter (`DETECT_DEBOUNCE = 3`).
- Implementation plan document.
- Change-report PDF.

4.2 Changed

- `setJoints()` signature: now 6 joints + gripper.
- Ultrasonic threshold: `dist <= 10.0 cm` (was `<= 8.0`).
- PCA9685 init standardized across all sketches.
- Servo model references: MG995 (per report) instead of MG996R/MG90S mix in architecture doc.
- Voltage regulator: XL4016.
- README status: Completed (v1.1).
- Sensor integration: gripper-mounted note.
- Color test logic: Yellow takes priority when R & G strong and B weak.

4.3 Removed

- None (no files deleted). Unused `USMIN`/`USMAX` constants retained for reference.

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5. Risk & Mitigation

Risk	Mitigation
No Arduino CLI available to compile	Manual syntax review; keep changes minimal and consistent
Yellow misclassification under ambient light	Add shielding & white-balance routine (Phase 5)
6th servo mechanical interference	Calibrate via `joint_calibration` before full runs
Base servo range (150-600) for 4 bins	Bin targets spaced: 200/375/550/590

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6. Acceptance Criteria

- All sketches compile-conceptually consistent (standardized init, channel defs).
- FSM supports 4 colors and 6 joints.
- Documentation matches report (6 MG995, XL4016, gripper-mounted sensor, 4 bins).
- Test records include 6 channels and Yellow.
- Implementation plan and change-report PDF delivered.

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7. Deliverables

1. Updated repository (firmware + docs + tests).
2. `IMPLEMENTATION_PLAN.md` (this file).
3. `docs/change_report.pdf` (change demonstration).
4. Git commit + push to `origin/main`.